Shashank Yadav Kalki

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PROFESSIONAL SUMMARY

Machine Learning postgraduate from University College Cork with research experience in statistical modelling, dynamical systems, and applied machine learning. Conducted MSc dissertation research investigating reliability, failure modes, and emergent behaviour in reservoir computing systems. Experienced in designing computational experiments, analysing large experimental datasets, and developing end-to-end machine learning pipelines using Python and SQL. Strong foundation in statistical learning, model evaluation, optimization, and data-driven problem solving with an interest in building scalable machine learning solutions.

TECHNICAL SKILLS

Programming: Python,NumPy, Pandas, Scikit-learn, SQL, R
Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Time Series Analysis
Deep Learning: Neural Networks, CNNs, RNN/LSTM, Reservoir Computing
Mathematics: Statistical Modelling, Optimization, Numerical Simulation, Dynamical Systems, Probability
Data Analysis: Statistical Inference, Data Cleaning, Error Analysis, Validation
Tools: FastAPI, Git/GitHub, Matplotlib, Seaborn, Power BI

RESEARCH EXPERIENCE

Machine Learning Research Assistant (MSc Dissertation) **May 2025 – Sept 2025**
University College Cork, Ireland | Supervisor: Dr. Andrew Flynn

- Designed and executed computational experiments involving parameter sweeps across reservoir architectures to investigate emergent behaviour in recurrent neural systems.
- Developed reproducible Python workflows for experimentation, statistical analysis, visualization, and model evaluation.
- Investigated reliability, stability, and failure modes of machine learning models through quantitative analysis.
- Applied numerical simulation, statistical modelling, and dynamical systems theory to evaluate prediction behaviour.
- Authored a 50-page MSc dissertation titled *Confabulation Dynamics in a Reservoir Computer*.

EDUCATION

University College Cork **Cork, Ireland**
M.Sc. Mathematical Modelling & Machine Learning Sept 2024 – Mar 2026
Relevant Coursework: Machine Learning, Deep Learning, Scientific Computing, Numerical Methods, Nonlinear Dynamics.

Mahatma Gandhi Institute of Technology **Hyderabad, India**
B.Tech Computer Science (Data Science) 2020 – 2024

PROJECTS

Product Recommendation System using Collaborative Filtering **Aug 2025 – Sept 2025**
Python | Surprise | Scikit-learn | Recommendation Systems

- Designed and implemented a personalized recommendation engine using collaborative filtering techniques.
- Performed exploratory data analysis, feature engineering, and similarity modelling to capture user preferences.
- Developed user-based and item-based recommendation models and evaluated ranking performance using Precision@K, Recall@K, Mean Average Precision (MAP), and NDCG.
- Optimized recommendation quality through hyperparameter tuning and comparative evaluation, producing scalable and reproducible experimentation workflows.

Heart Disease Prediction – End-to-End ML Pipeline **Jan 2025 – Mar 2025**
Python | Scikit-learn | FastAPI

- Built an end-to-end ML pipeline including preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation.
- Compared supervised learning algorithms using Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- Performed feature selection and data validation to improve robustness.
- Developed FastAPI endpoints for model inference.

Human Activity Recognition **Oct 2024 – Jan 2025**
Python | PCA | Time Series

- Engineered time-series features from wearable sensor datasets.
- Applied PCA to improve computational efficiency while preserving predictive information.
- Evaluated classification models and analysed error patterns.
- Compared multiple classification models and analysed prediction errors to improve generalization.

CERTIFICATIONS

- Data Science for Engineers – IIT Madras
- Natural Language Processing – IIT Kharagpur
- Data Analysis with Python – FreeCodeCamp
- Social Network Analysis – IIT Ropar
- Prompt Engineering with OpenAI – Columbia University
- Generative AI Masterclass – Udemy